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Azure AI Foundry Architecture Whitepaper

A technical deep dive into Azure AI Foundry: Model Catalog, Prompt Flow LLMOps, Semantic Kernel orchestration patterns, Azure AI Search RAG pipelines, enterprise security configuration, and phased rollout framework.

Architecture	Prompt Flow	Semantic Kernel	RAG	Entra ID

Produced by **Kovil AI Engineering Team** · kovil.ai · June 2026

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01 / Overview

Azure AI Foundry Stack — Overview

Azure AI Foundry is a layered platform. Understanding which component belongs at which layer — and which is optional — is the first architecture decision.

Model Layer **Azure AI Model Catalog (1,600+ models)**	GPT-4o, GPT-4o-mini (Azure OpenAI), Phi-3, Llama 3.3, Mistral, Command R+ — available as managed endpoints with SLAs, RBAC, and Private Endpoint support. Model selection is the first architecture decision.
Orchestration Layer **Prompt Flow / Semantic Kernel**	Two orchestration options. Prompt Flow: DAG-based visual LLMOps with built-in evaluation. Semantic Kernel: code-first (Python/C#) agent framework with plugin architecture. Choose based on team profile and orchestration complexity.
Retrieval Layer **Azure AI Search**	Enterprise vector + keyword search for RAG pipelines. Supports Integrated Vectorisation (automatic embedding during indexing), semantic ranking, hybrid retrieval, and document-level security trimming.
Safety Layer **Azure AI Content Safety**	Content moderation for prompts and completions. Configurable severity thresholds per category (hate, violence, sexual, self-harm). Groundedness detection for RAG agents to flag hallucinations.
Platform Layer **Azure AI Foundry Hub & Projects**	Shared infrastructure layer (compute, storage, key vault, connections) + per-use-case project containers. The governance and billing boundary for all AI Foundry workloads.
Governance Layer **Entra ID, Azure Policy, Azure Monitor**	Identity (Managed Identity, RBAC), network (Private Endpoints, VNet Integration), compliance (Azure Policy, CMK, diagnostic logs). Applied at the Hub resource group level.

02 / Models

Model Catalog — Foundation Model Selection

Model	Best For	Context	Deploy Via
GPT-4o	Complex reasoning, multi-modal, production ag	128K tokens	Azure OpenAI
GPT-4o-mini	High-volume, latency-sensitive, cost-optimised	128K tokens	Azure OpenAI
o1 / o1-mini	Advanced reasoning, complex analysis tasks	200K tokens	Azure OpenAI
Phi-3.5-mini	Edge deployment, low-latency, small model	128K tokens	Model Catalog
Llama 3.3 70B	Data sovereignty, fine-tuning, cost control	128K tokens	Model Catalog
Mistral Large	EU data residency, multilingual	128K tokens	Model Catalog
text-embedding-3-large	Production RAG embeddings (3072 dims)	N/A	Azure OpenAI

Azure OpenAI models have enterprise SLAs, RBAC, Private Endpoint support, and Content Safety integration. Model Catalog models (Llama, Mistral, Phi) are deployed on managed compute in your Azure subscription — no data leaves your environment.

03 / LLMOps

Prompt Flow — LLMOps Architecture

Prompt Flow is Azure AI Foundry's LLMOps platform — a DAG-based visual workflow for building, evaluating, and deploying LLM pipelines. It is the recommended orchestration layer for teams that prioritise operational consistency, evaluation-driven development, and CI/CD integration over raw orchestration flexibility.

Standard Flow vs Chat Flow vs Evaluation Flow

Standard: batch processing pipelines. Chat: multi-turn conversation agents with history management. Evaluation: quality measurement flows that assess other flows. Build your agent as a Chat Flow; evaluate it with an Evaluation Flow.

Node Types

LLM nodes (Azure OpenAI calls), Python nodes (custom code, API calls, data processing), Prompt nodes (template rendering). Complex agents combine multiple node types in a directed acyclic graph.

Connection Management

Prompt Flow uses Azure AI Foundry Connections to securely store API keys and endpoints (Azure OpenAI, Azure AI Search, custom APIs). Connections are Hub-level resources — shared across all Projects in the Hub.

Evaluation Framework

Prompt Flow includes built-in evaluators: groundedness (does the answer come from the context?), relevance (is the answer relevant to the question?), coherence, fluency, and similarity. Run evaluations on every significant prompt change.

CI/CD Integration

Prompt Flow exports flows as Python packages with a YAML manifest. Integrate into Azure DevOps or GitHub Actions pipelines: test → evaluate → deploy to Azure ML online endpoint. Automated evaluation gates prevent quality regressions.

Tracing & Monitoring

Prompt Flow integrates with Azure Monitor Application Insights for distributed tracing — every node execution, LLM call, token count, and latency is captured. Set up dashboards for token consumption, latency p95, and evaluation score trends.

04 / Orchestration

Semantic Kernel Orchestration Patterns

Semantic Kernel (SK) is Microsoft's open-source agent orchestration framework — Python and C# — for building complex, multi-step AI agents with full code control. Use SK when Prompt Flow's DAG model is too restrictive for your orchestration needs.

Pattern 1: Single Agent + Plugin Architecture

One SK agent with multiple plugins — each plugin is a Python or C# function decorated with @kernel_function. The agent selects the right plugin based on the user's intent using the LLM planner. Use for: self-contained agents with a defined tool set (CRM lookup, calendar booking, document search).

Pattern 2: Multi-Agent Orchestration

Multiple SK agents, each with a specialised role, coordinated by an orchestrator agent. The orchestrator decomposes complex tasks and delegates to specialist agents. Use for: complex research agents, multi-system data aggregation, agentic process automation.

Pattern 3: SK + Azure AI Search RAG

SK agent uses a custom plugin that wraps Azure AI Search — retrieving relevant context before every LLM call. The SK planner decides when to retrieve and what to retrieve based on query analysis. Use for: knowledge-grounded agents where retrieval must be selective, not always-on.

Pattern 4: SK + Persistent Memory

SK agent stores conversation summaries and user preferences in Azure Cosmos DB or Azure Cache for Redis. On subsequent sessions, the agent retrieves relevant memory and incorporates it into the system prompt. Use for: personalised agents that improve over multiple sessions.

Pattern 5: SK Agent + Azure Logic Apps

SK agent triggers Azure Logic Apps workflows as tool calls — accessing 400+ connectors (Salesforce, Dynamics, SAP, Jira, etc.) without custom integration code. Use for: enterprise process automation agents that need broad system connectivity.

05 / Retrieval

Azure AI Search RAG Pipeline

Azure AI Search with Integrated Vectorisation is the recommended RAG backbone for Azure AI Foundry agents. Understanding each pipeline stage determines index quality and agent accuracy.

1. Data Ingestion

Supported sources: Azure Blob Storage (PDF, DOCX, HTML, TXT, JSON), Azure SQL, Cosmos DB, SharePoint Online, and OneLake (Fabric). Use built-in indexers for managed crawling or the push API for real-time updates.

2. Integrated Vectorisation

Enable Integrated Vectorisation to have Azure AI Search automatically call your Azure OpenAI embedding model during indexing — no separate embedding pipeline required. Embeddings are stored in the same index as the text content.

3. Chunking Configuration

Default chunk size is 512 tokens with 20% overlap. For technical documentation: 256–384 tokens improves precision. For narrative content: 512–768 tokens preserves context. Test both against a sample query set.

4. Semantic Ranking

Azure AI Search Semantic Ranker re-ranks the top-50 BM25 results using a cross-encoder model. Enable semantic ranking for all production deployments — it improves answer accuracy by 20–35% on natural language queries at minimal additional cost.

5. Hybrid Retrieval

Combine keyword (BM25) and vector search with RRF (Reciprocal Rank Fusion) for best overall retrieval quality. Pure vector search misses exact-match queries; pure keyword search misses semantic paraphrases. Hybrid retrieval handles both.

6. Groundedness Detection

Azure AI Content Safety includes a groundedness detection API — verifying that agent responses are grounded in the retrieved context. Enable for production RAG agents to catch hallucinations before they reach users.

06 / Security

Enterprise Security Configuration

Managed Identity (No Secrets Architecture)

All Azure AI Foundry workloads should use User-Assigned Managed Identity for authentication to Azure OpenAI, Azure AI Search, Azure Key Vault, and Azure Storage. No API keys in code, no rotation risk, full RBAC control.

Private Endpoints

In production, all Azure AI Foundry resources communicate via Private Endpoints — eliminating public internet exposure. Private DNS zones must be configured for each service. Test connectivity from your compute before go-live.

Customer-Managed Keys (CMK)

Encrypt data at rest in Azure OpenAI and Azure AI Search with keys stored in Azure Key Vault. Required for HIPAA, FedRAMP High, ISO 27001 audits. Configure before any production data is ingested — retroactive CMK application is complex.

Entra ID Conditional Access

Apply Conditional Access policies to Azure AI Foundry portal access: require MFA, compliant devices, and named locations for all administrative access. Block access from non-corporate networks for sensitive AI workloads.

Network Isolation

Configure Azure AI Foundry Hub with Public Network Access disabled. Developers access the Hub via Azure Bastion or VPN. CI/CD pipelines deploy via Private Endpoint from a self-hosted runner in the VNet.

Azure Policy — AI Guardrails

Enforce: require Private Endpoints, require Managed Identity, restrict model deployments to approved model list, require content safety configuration, require diagnostic logging. Apply at the Management Group level for consistent enforcement.

07 / Rollout

Phased Production Rollout Framework

Phase 1: Internal Alpha (Days 1–10)

Deploy to Azure AI Foundry staging environment. Internal team of 5–10 testers run 40+ defined test queries covering success, edge case, and out-of-scope scenarios. Configure Application Insights tracing. Set acceptance criteria: >80% groundedness score, <4s p95 latency, 0% safety filter false positives on legitimate queries.

> **Gate:** Groundedness >80%, latency <4s p95, 0 safety false positives on test set.

Phase 2: Limited Beta (Days 11–28)

Deploy to production Azure AI Foundry project. Route to 10–30 selected internal users. Review Application Insights traces daily — identify top 5 failure patterns. Iterate on prompt templates and retrieval configuration. Do not expand user base until failure rate is below 8%.

> **Gate:** 10–30 beta users, daily trace review, failure rate <8% before expanding.

Phase 3: Canary Rollout (Days 29–42)

Route 10–20% of production traffic via Azure API Management weighted routing policy. Monitor groundedness rate, latency, error rate in real time via Azure Monitor dashboards. Set automated alerts: groundedness <70% triggers review; error rate >3% triggers rollback. Keep manual process running in parallel.

> **Gate:** 10–20% traffic via APIM. Automated rollback threshold defined and tested.

Phase 4: Full Production (Day 43+)

Ramp to 100% traffic. Establish weekly evaluation runs using Prompt Flow evaluation flows on a held-out query set. Schedule monthly prompt and index updates. Review Application Insights token consumption and latency trends weekly.

> **Gate:** Weekly evaluation runs. Monthly optimisation cycle. Monitoring dashboards live.

08 / Checklist

Architecture Decision Checklist

#	Decision	Your Answer
1	Model selection GPT-4o vs GPT-4o-mini vs Phi-3 vs Llama? Based on latency, cost, and accuracy requirements.	
2	Orchestration layer Prompt Flow (LLMOps-first) vs Semantic Kernel (code-first) vs Copilot Studio (M365)?	
3	RAG configuration Chunk size, overlap, semantic ranking, hybrid retrieval weights — tested against sample queries?	
4	Embedding model text-embedding-3-large (quality) vs text-embedding-3-small (cost)?	
5	Data freshness Indexer schedule defined. Push API configured for real-time updates (if needed)?	
6	Security trimming Document-level ACLs configured in Azure AI Search for sensitive content?	
7	Identity User-assigned Managed Identity configured for all resource connections?	
8	Network isolation Private Endpoints for Azure OpenAI and Azure AI Search (production)?	
9	CMK Customer-Managed Keys configured (if required by compliance framework)?	
10	Content safety Azure AI Content Safety thresholds set. Groundedness detection enabled?	
11	Monitoring App Insights tracing live. Azure Monitor dashboards configured. Alerts set?	
12	Evaluation Prompt Flow evaluation flow built. CI/CD evaluation gate in place?	

Need help completing this checklist? Kovil AI runs a 2-week Azure AI Foundry Strategy & Readiness engagement that works through every architecture decision and produces a written blueprint. Contact us at kovil.ai.